Fe-C Problem 6 | Problems T4_PhaseDiagrams II.pdf #6 | Fe-C · hypoeutectic cast iron

Hypoeutectic Cast Iron from its Ferrite Fraction

Original statement · Problems T4_PhaseDiagrams II.pdf, p. 2

6.- Consider a steel with a content of C < 4.3% at (727−ΔT)ºC. In these conditions, χ_α = 0.47. Indicate the mass fraction of the phases at (1147+ΔT)ºC and (1147−ΔT)ºC.

Official Solution: T1: w_γ **= 0.32,** χ_L **= 0.68; T2:** w_γ **= 0.68,** χ_{Fe_3C} **= 0.32,** w_E **= 0.68,** $\chi_{\gamma E}$ **= 0.36,** w_{Fe_3C} **= 0.32**

PHASE 1 · HYPOTHESES & PARAMETERS

1. Equilibrium, Fe-Fe3C diagram [Session 9 Slide 4]: α 0.022, γ 2.14 (at 1147 °C), eutectic 4.30, Fe3C 6.67 % C.
2. Unknown first: C_0, from the ferrite fraction just below 727 °C.
3. The statement says "steel", but the result is above 2.14 % C, so the alloy is a **hypoeutectic cast iron** (it has ledeburite).

PHASE 2 · FRAMES & STRATEGY

Frame: the Fe-Fe3C diagram (wt% C, °C), with tie lines at 727 − ΔT (α-Fe3C), 1147 + ΔT (γ-L) and 1147 − ΔT (γ-Fe3C). Step 1 solves the lever rule for C_0; steps 2-3 apply it at 1147 ± ΔT; microconstituents as in Session 7 Slide 32.

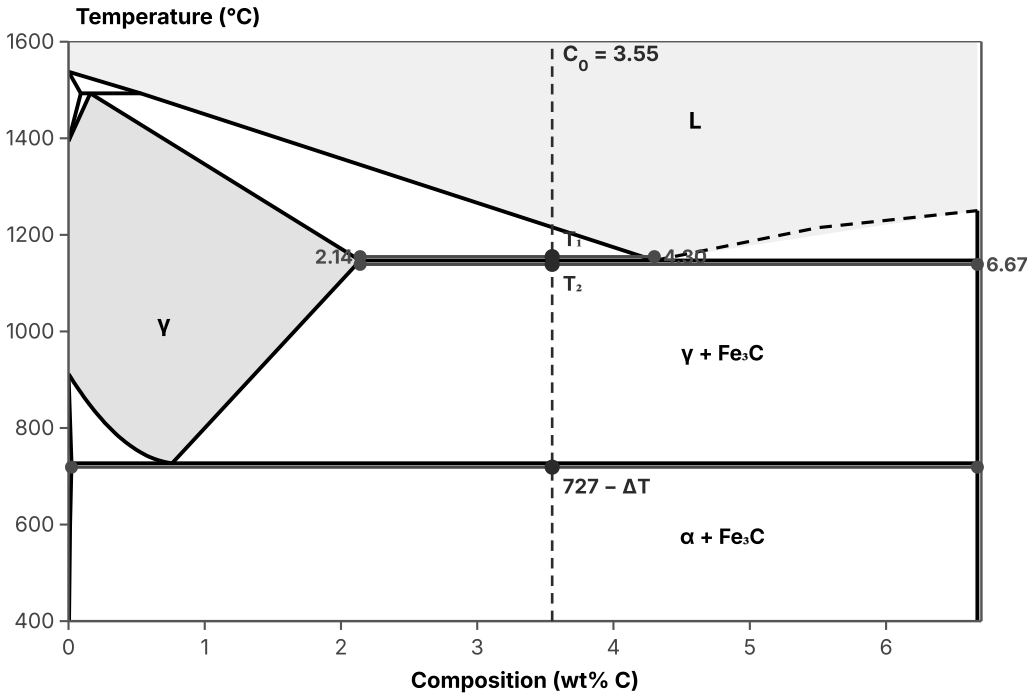


Figure 34. The alloy with 47 % α just below 727 °C lies at about 3.55 % C: γ + L above 1147 °C, γ + Fe₃C below it.

PHASE 3 · STEP-BY-STEP DERIVATION

1. Composition

$$W_\alpha = \frac{C_{\mathrm{Fe_3C}} - C_0}{C_{\mathrm{Fe_3C}} - C_\alpha} \;\Rightarrow\; C_0 = C_{\mathrm{Fe_3C}} - W_\alpha\,(C_{\mathrm{Fe_3C}} - C_\alpha) = 6.67 - 0.47 \times 6.648 = 6.67 - 3.125 = 3.545 \approx 3.55\ \%\,\mathrm{C}$$

2. T_1 = 1147 + ΔT (γ 2.14 / L 4.30; length 2.16)

$$W_L = \frac{C_0 - C_\gamma}{C_L - C_\gamma} = \frac{3.545 - 2.14}{2.16} = \frac{1.405}{2.16} = 0.650 \qquad W_\gamma = \frac{4.30 - 3.545}{2.16} = \frac{0.755}{2.16} = 0.350$$

3. T_2 = 1147 – ΔT (γ 2.14 / Fe₃C 6.67; length 4.53)

$$W_{\mathrm{Fe_3C}} = \frac{3.545 - 2.14}{4.53} = \frac{1.405}{4.53} = 0.310 \qquad W_\gamma = \frac{6.67 - 3.545}{4.53} = \frac{3.125}{4.53} = 0.690$$

Microconstituents at T_2: primary austenite = γ at T_1 = 0.350; ledeburite = L at T_1 = 0.650, made of γ $0.690 - 0.350 = 0.340$ and Fe₃C 0.310.

PHASE 4 · CHECKS & INTERPRETATION

- Ledeburite make-up: $0.340/0.650 = 0.523$ γ, as in Worked example 14 .
- Mass balance at T_2: $0.690 \times 2.14 + 0.310 \times 6.67 = 1.48 + 2.07 = 3.55$ % C .
- **Limiting cases** (fractions dimensionless, compositions in wt% C): the ferrite fraction $(6.67 - C_0)/6.648$ is 1 at $C_0 = 0.022$ and 0 at $C_0 = 6.67$; 0.47 lies between, so C_0 lies between those limits, here inside the cast-iron range.

Check against the official solution

The method matches the worked page of Solution problems Phase diagrams II.pdf, p. 12.

DISCREPANCY WITH THE OFFICIAL SOLUTION

The worked page rounds the composition to $C_0 = 3.6$ % C before going on (with 6.7 % C it is 3.56) and so gives 0.68/0.32 at T_1, 0.32/0.68 at T_2, primary γ 0.32 and eutectic γ 0.36. With the unrounded $C_0 = 3.545$ % C the values are 0.650/0.350, 0.310/0.690, primary γ 0.350 and eutectic γ 0.340; the computed values are kept. The page also writes the formula for χ_L with the labels of the α-Fe₃C tie line; its numbers, $(3.6 - 2.14)/(4.3 - 2.14)$, are those of the γ-L tie line used here.